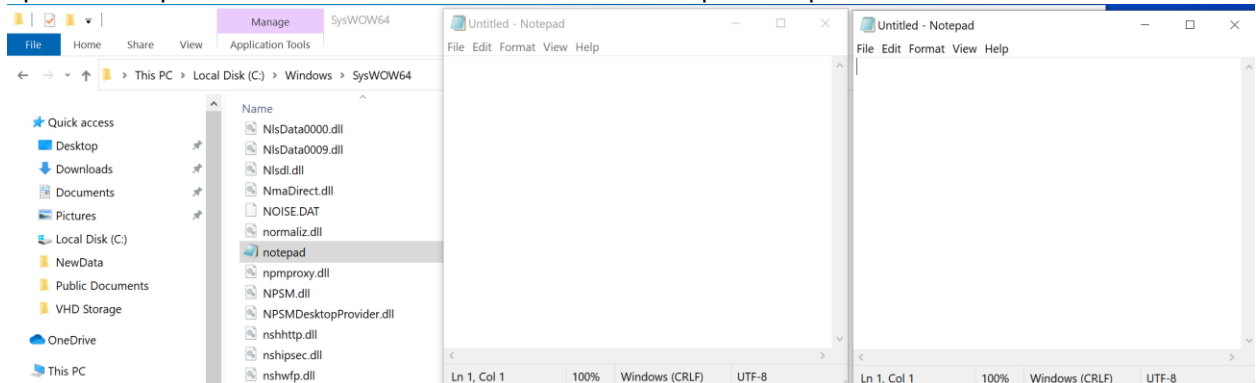


Activity 9-1: Identifying Win32 and Win64 Apps

1. Opened the notepad by searching on the task bar and go to C:\Windows\SysWOW64 and opened notepad. There was no difference between both open notepad windows.



2. Opened the Task manager and viewed both notepads. One of notepad apps has (32 bit) appended to the name. The text (32 bit) identifies the Win32 process.

Task Manager

File Options View

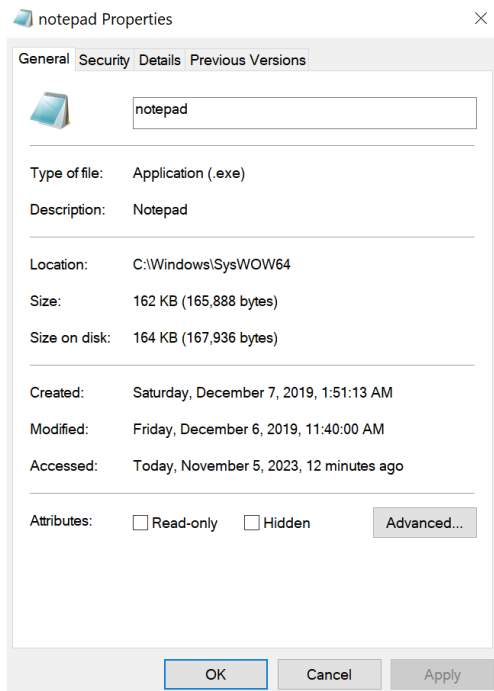
Processes Performance App history Startup Users Details Services

Name	Status	100% CPU	60% Memory	80% Disk	0% Network	Power usage	Power usage t...
> Service Host: DNS Client		0%	1.4 MB	0 MB/s	0 Mbps	Very low	Very low
> Notepad (32 bit)		0%	1.4 MB	0 MB/s	0 Mbps	Very low	Very low
> Spooler SubSystem App		0%	1.3 MB	0 MB/s	0 Mbps	Very low	Very low
> Microsoft Network Realtime Ins...		0%	1.3 MB	0 MB/s	0 Mbps	Very low	Very low
> Service Host: User Profile Service		0%	1.2 MB	0 MB/s	0 Mbps	Very low	Very low
> wsappx		0%	1.2 MB	0 MB/s	0 Mbps	Very low	Very low
> Service Host: Time Broker		0%	1.2 MB	0 MB/s	0 Mbps	Very low	Very low
> Touch Keyboard and Handwritin...		0%	1.2 MB	0 MB/s	0 Mbps	Very low	Very low
> Notepad		0%	1.2 MB	0 MB/s	0 Mbps	Very low	Very low
> Service Host: Local Service (Net...		0%	1.1 MB	0 MB/s	0 Mbps	Very low	Very low
> Service Host: IP Helper		0%	1.1 MB	0 MB/s	0 Mbps	Very low	Very low

⏮ Fewer details

End task

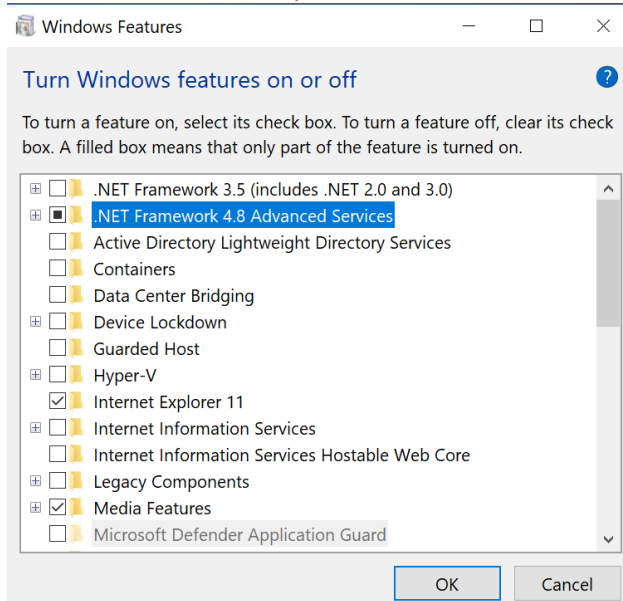
3. Opened the properties of notepad (32 bit) and noticed that location is C:\Windows\SysWOW64.



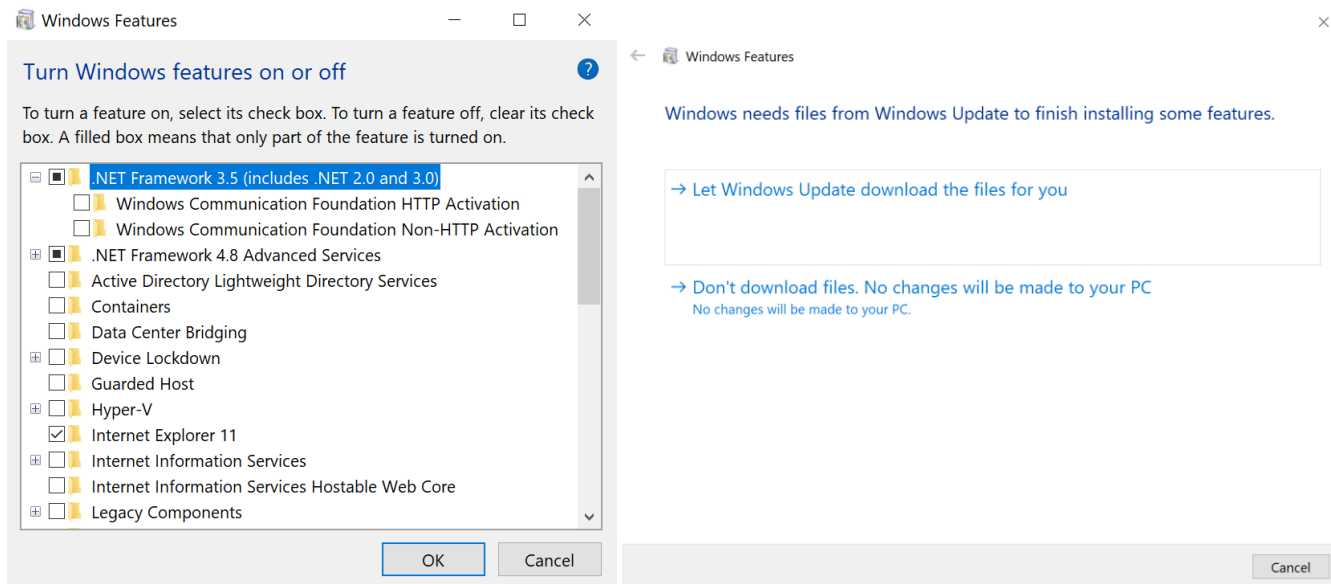
4. Closed all open windows.

Activity 9-2: Installing .NET Framework 3.5

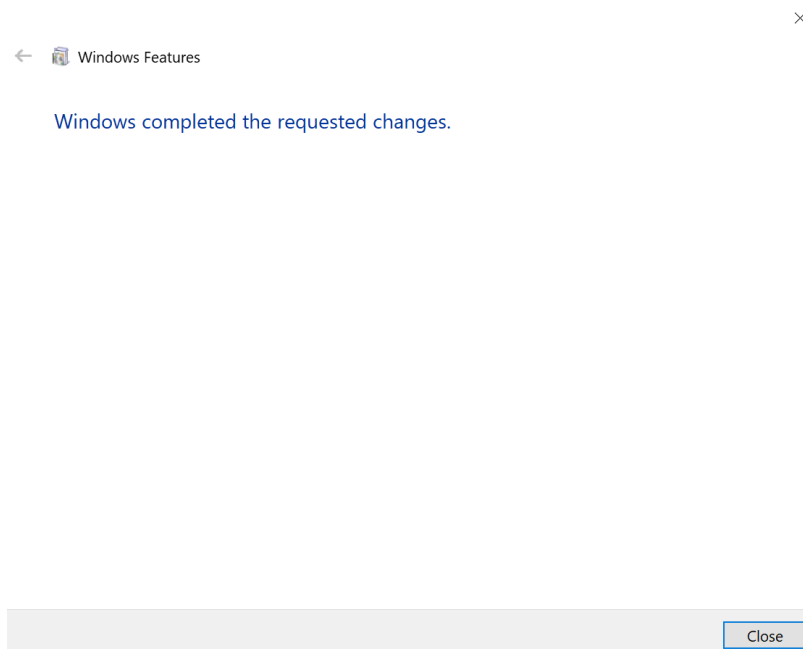
1. Searched for feature and select Turn Windows features on or off and noticed that .NET Framework 4.8 is already installed.



- Clicked .Net Framework 3.5 (includes .NET 2.0 and 3.0) and clicked Download files from Windows Update.

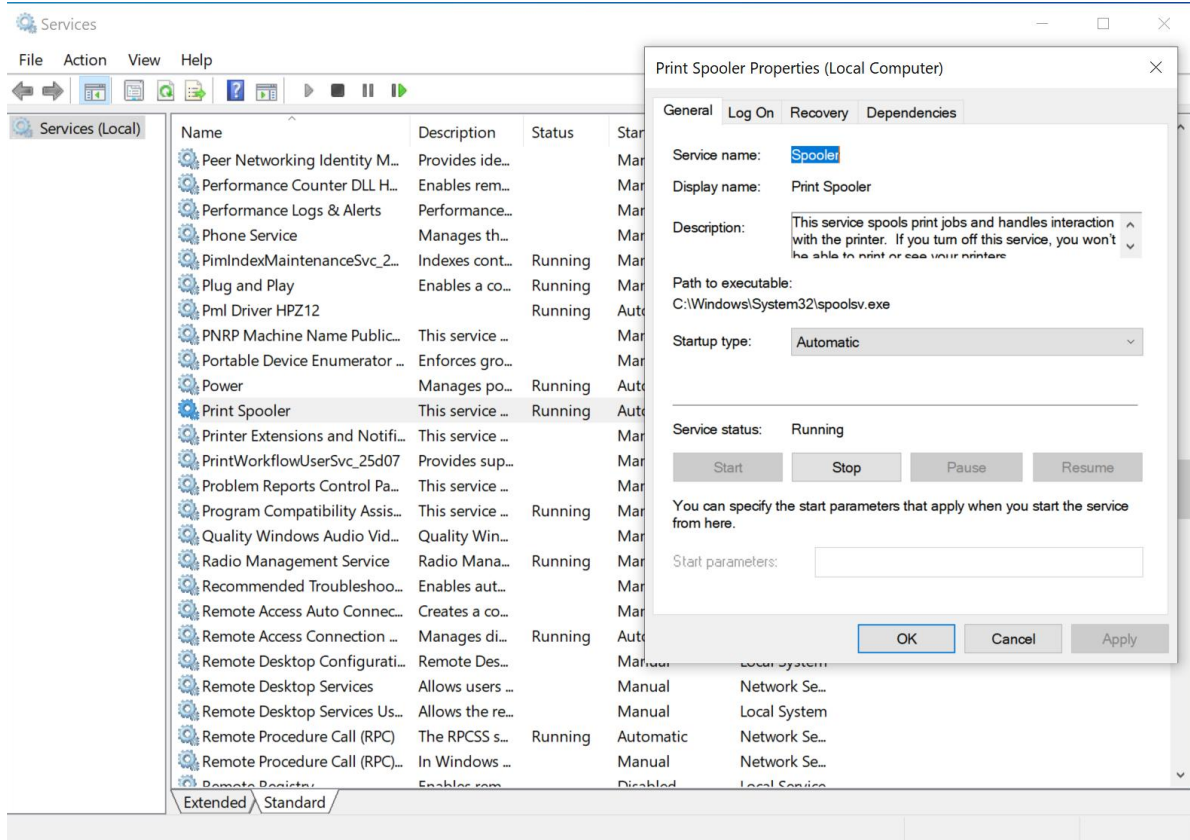


- When windows has completed the requested changes, closed the window.

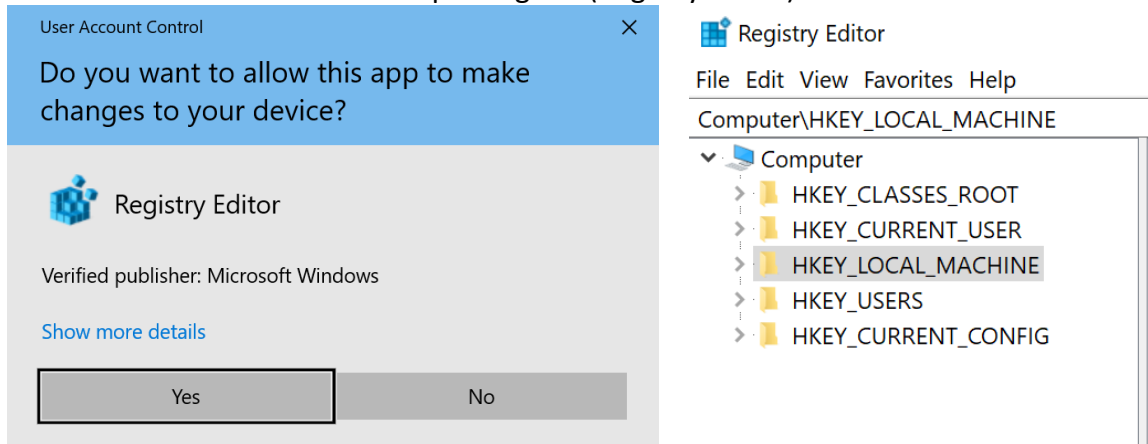


Activity 9-3: Using Regedit to Back Up and Modify the Registry

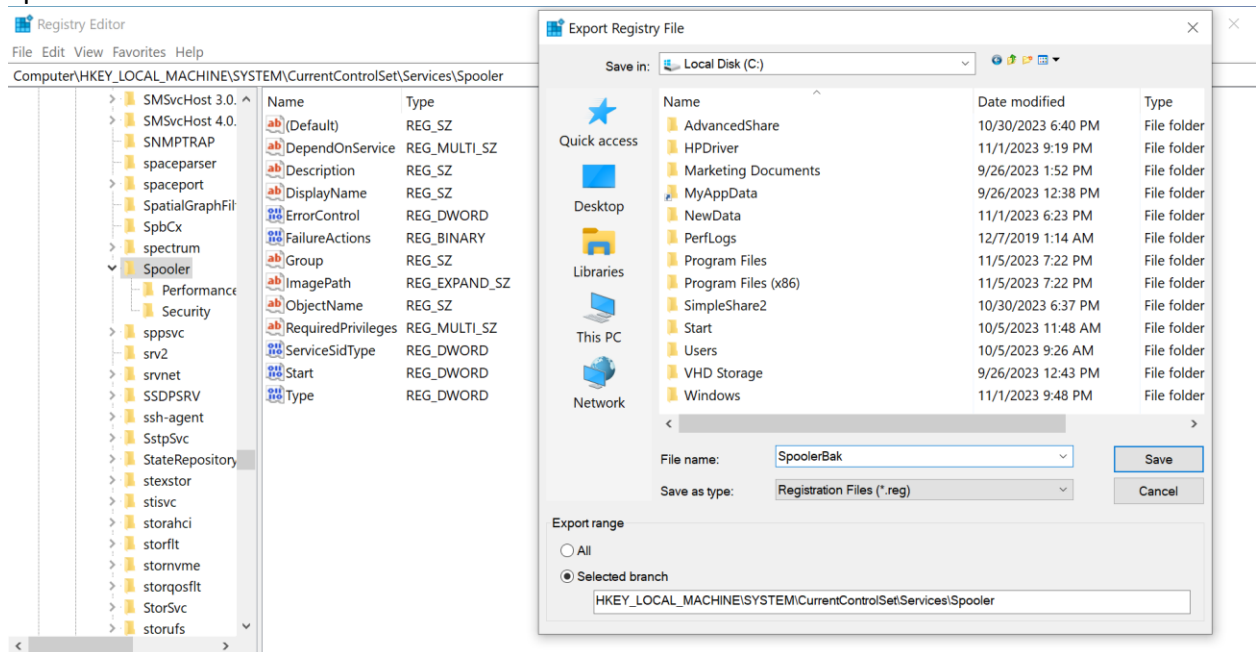
1. Opened Services and double-clicked Printer Spooler services. Read the information about the service and noticed that startup type is Automatic.



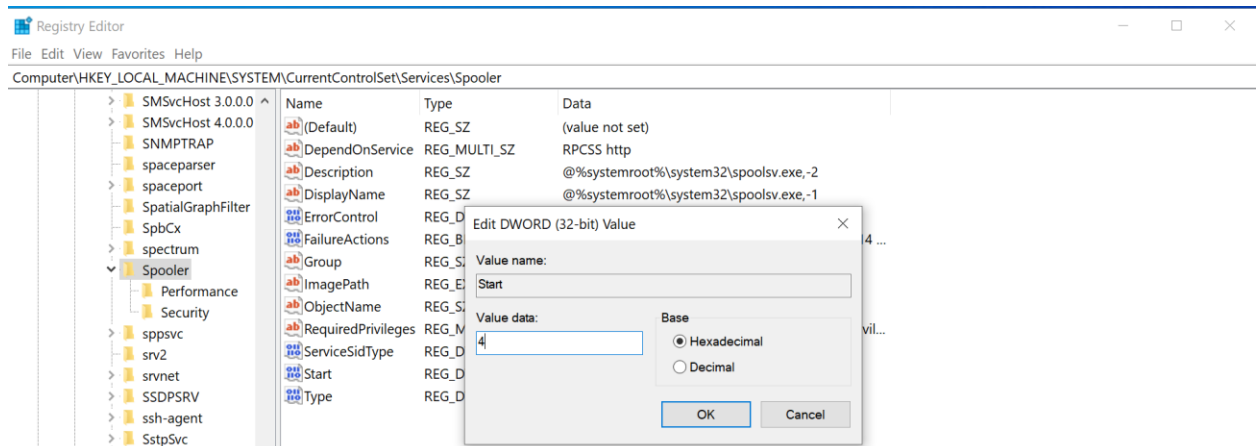
2. Searched in the search bar and open regedit (Registry Editor).



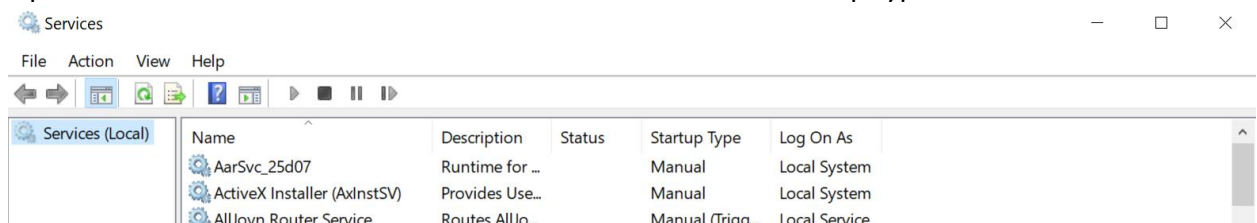
- Opened HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Spooler and right clicked spooler and clicked Export. Selected Local Disk C: and Saved File name as SpoolerBak.



- Double Clicked Start in the Registry Editor and changed Value data box to value 4 and clicked Ok.



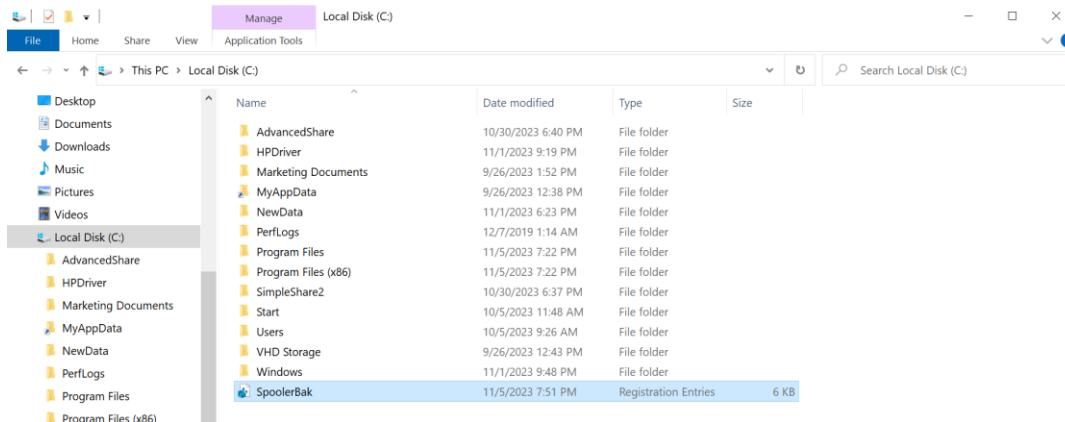
- Opened the Service window and refreshed it. Noticed that Startup type is disabled.



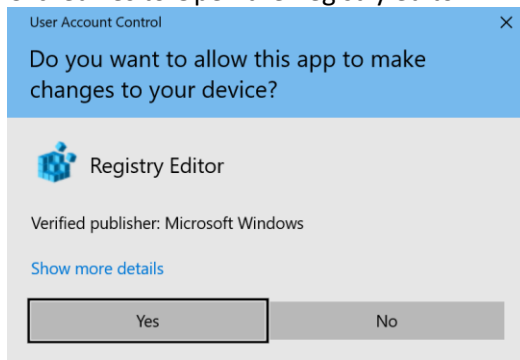
- In the Register Editor, collapsed all the Hives and closed the Registry editor.

Activity 9-4: Restoring a Registry Backup

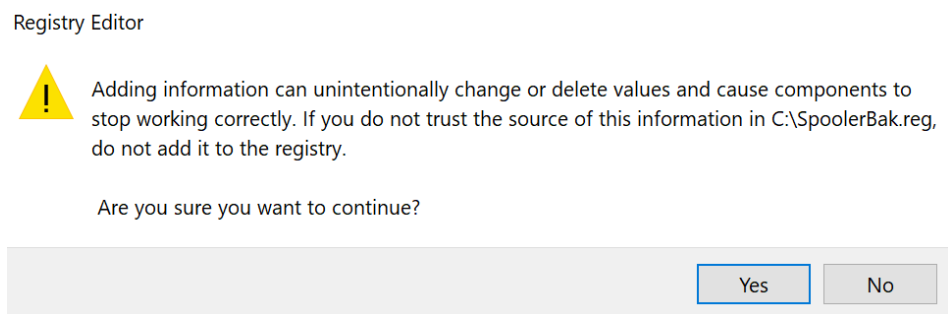
1. Opened the File Explorer and opened C:\SpoolerBak.



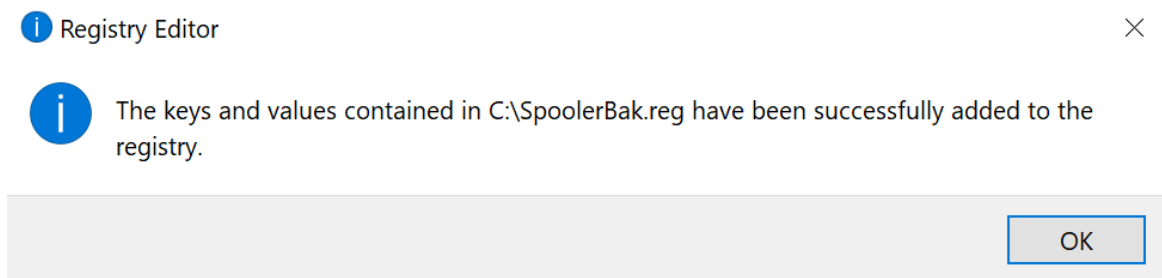
2. Clicked Yes to Open the Registry editor.



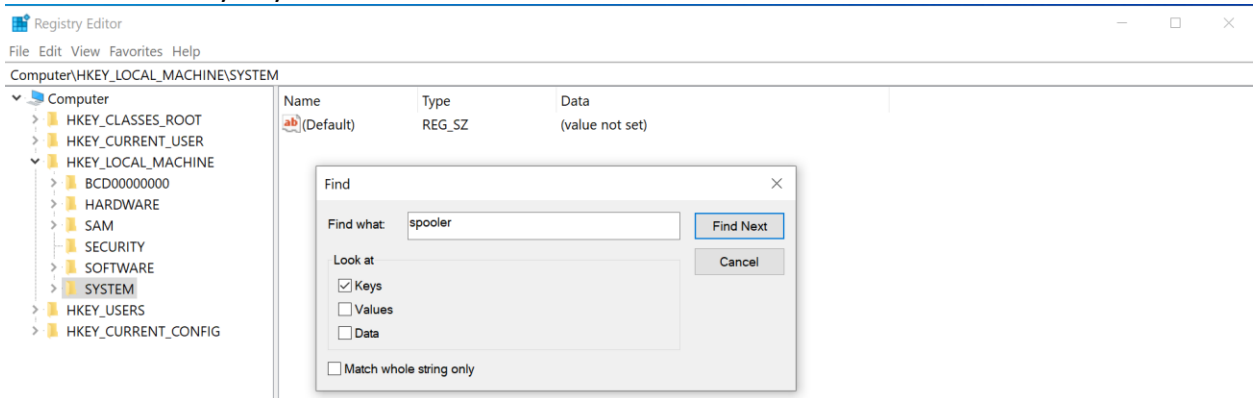
3. In the Registry Editor window, clicked Yes to continue.



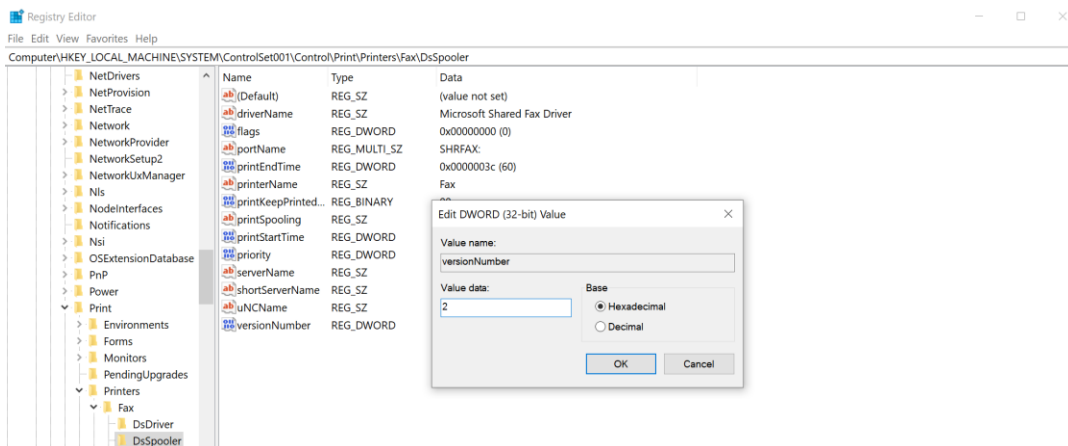
4. Acknowledged the successful import and closed the File Explorer.



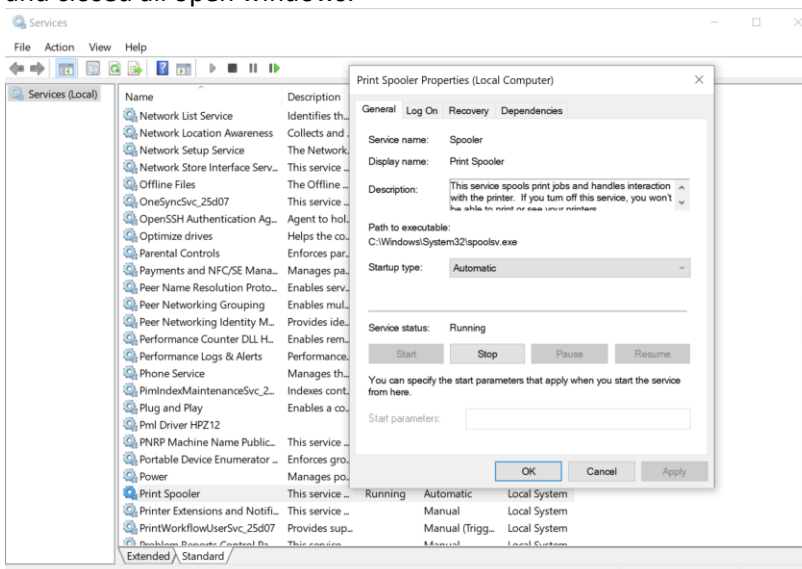
5. Searched in the search bar and open regedit (Registry Editor). Selected HKEY_LOCAL_MACHINE\SYSTEM. Then clicked Edit and selected Find and typed spooler and selected only Keys in check boxes.



6. Read the Start value and verified that it has been set back to 2.

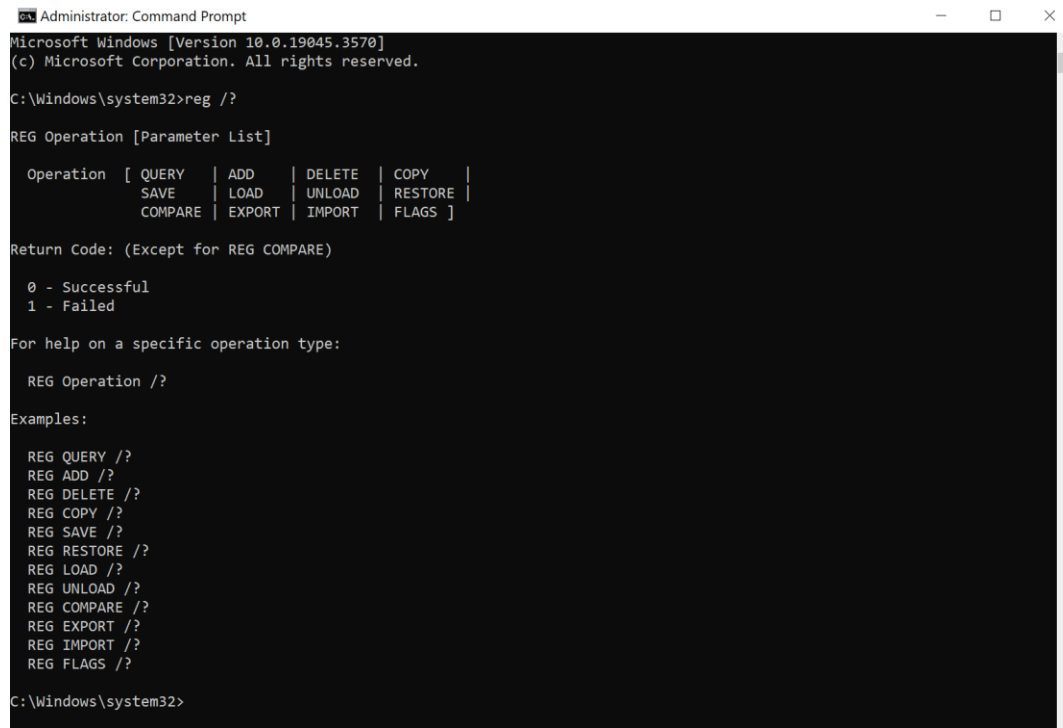


7. Opened Services and clicked Print Spooler service and verified that the Startup Type is Automatic and closed all open windows.



Activity 9-5: Using Reg.exe

1. Opened the command prompt (Admin) and typed reg /? Command to view the general help information.



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.3570]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>reg /?

REG Operation [Parameter List]

Operation [ QUERY | ADD | DELETE | COPY |
           SAVE | LOAD | UNLOAD | RESTORE |
           COMPARE | EXPORT | IMPORT | FLAGS ]

Return Code: (Except for REG COMPARE)

0 - Successful
1 - Failed

For help on a specific operation type:

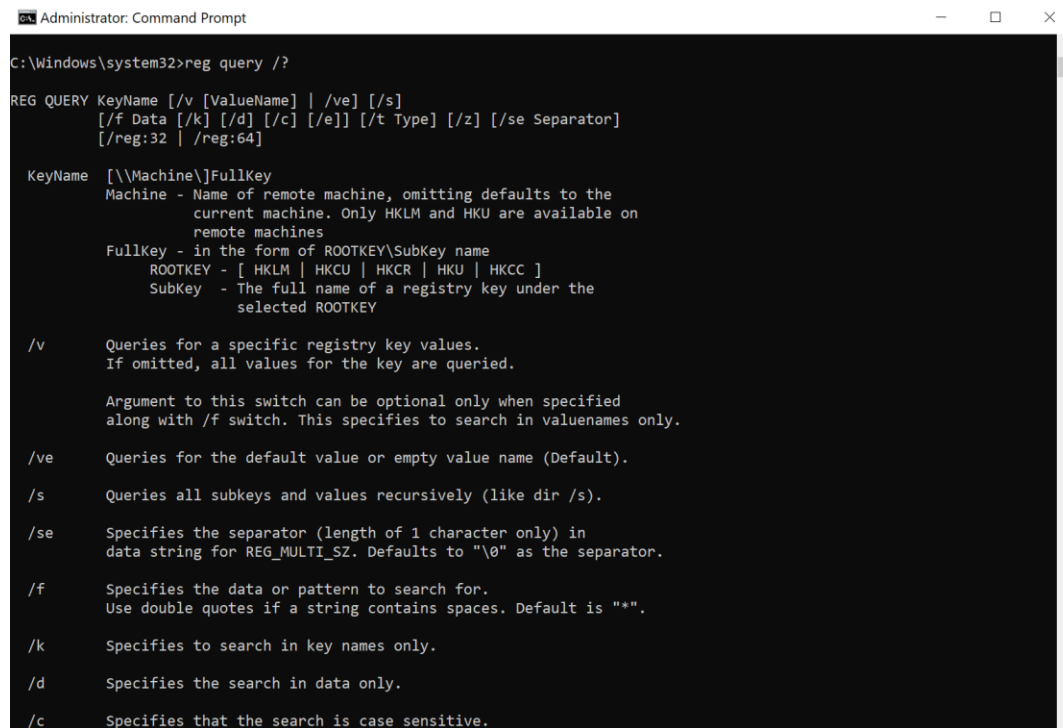
REG Operation /?

Examples:

REG QUERY /?
REG ADD /?
REG DELETE /?
REG COPY /?
REG SAVE /?
REG RESTORE /?
REG LOAD /?
REG UNLOAD /?
REG COMPARE /?
REG EXPORT /?
REG IMPORT /?
REG FLAGS /?

C:\Windows\system32>
```

2. Typed command reg query /? Command to view the query help information.



```
Administrator: Command Prompt

C:\Windows\system32>reg query /?

REG QUERY KeyName [/v [ValueName] | /ve] [/s]
           [/f Data [/k] [/d] [/c] [/e]] [/t Type] [/z] [/se Separator]
           [/reg:32 | /reg:64]

KeyName   [\\Machine\]FullKey
Machine   - Name of remote machine, omitting defaults to the
            current machine. Only HKLM and HKU are available on
            remote machines
FullKey    - in the form of ROOTKEY\SubKey name
            ROOTKEY - [ HKLM | HKCU | HKCR | HKU | HKCC ]
            SubKey   - The full name of a registry key under the
                        selected ROOTKEY

/v         Queries for a specific registry key values.
           If omitted, all values for the key are queried.

           Argument to this switch can be optional only when specified
           along with /f switch. This specifies to search in valuenames only.

/ve        Queries for the default value or empty value name (Default).

/s         Queries all subkeys and values recursively (like dir /s).

/se        Specifies the separator (length of 1 character only) in
           data string for REG_MULTI_SZ. Defaults to "\" as the separator.

/f         Specifies the data or pattern to search for.
           Use double quotes if a string contains spaces. Default is "*".

/k         Specifies to search in key names only.

/d         Specifies the search in data only.

/c         Specifies that the search is case sensitive.
```

3. Typed command `reg query HKLM\SYSTEM\CurrentControlSet\Services\Spooler` and pressed Enter. Verified that the start value is 0x2.

```
C:\Windows\system32>reg query HKLM\SYSTEM\CurrentControlSet\Services\Spooler

HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Spooler
    DependOnService    REG_MULTI_SZ    RPCSS\0http
    Description        REG_SZ          @%systemroot%\system32\spoolsv.exe,-2
    DisplayName        REG_SZ          @%systemroot%\system32\spoolsv.exe,-1
    ErrorControl        REG_DWORD       0x1
    FailureActions      REG_BINARY      100E0000000000000000000000300000014000000010000008813000001000000881300000000000000
0000
    Group              REG_SZ          SpoolerGroup
    ImagePath          REG_EXPAND_SZ   %SystemRoot%\System32\spoolsv.exe
    ObjectName         REG_SZ          LocalSystem
    RequiredPrivileges REG_MULTI_SZ    SeTcbPrivilege\0SeImpersonatePrivilege\0SeAuditPrivilege\0SeChangeNotifyPrivil
ege\0SeAssignPrimaryTokenPrivilege\0SeLoadDriverPrivilege
    ServiceSidType     REG_DWORD       0x1
    Start              REG_DWORD       0x2
    Type               REG_DWORD       0x110

HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Spooler\Performance
HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Spooler\Security

C:\Windows\system32>
```

4. Typed command `reg add /?` To view the add help information.

```
C:\Windows\system32>reg add /?
```

```
REG ADD KeyName [/v ValueName | /ve] [/t Type] [/s Separator] [/d Data] [/f]
[/reg:32 | /reg:64]
```

KeyName [\\Machine\]FullKey

Machine Name of remote machine - omitting defaults to the current machine. Only HKLM and HKU are available on remote machines.

FullKey ROOTKEY\SubKey

ROOTKEY [HKLM | HKCU | HKCR | HKU | HKCC]

SubKey The full name of a registry key under the selected ROOTKEY.

/v The value name, under the selected Key, to add.

/ve adds an empty value name (Default) for the key.

/t RegKey data types
[REG_SZ | REG_MULTI_SZ | REG_EXPAND_SZ |
 REG_DWORD | REG_QWORD | REG_BINARY | REG_NONE]
If omitted, REG_SZ is assumed.

/s Specify one character that you use as the separator in your data string for REG_MULTI_SZ. If omitted, use "\\0" as the separator.

/d The data to assign to the registry ValueName being added.

/f Force overwriting the existing registry entry without prompt.

/reg:32 Specifies the key should be accessed using the 32-bit registry view.

/reg:64 Specifies the key should be accessed using the 64-bit registry view.

Examples:

```
REG ADD \\ABC\HKLM\Software\MvCo
```

5. Typed command `reg add HKLM\SYSTEM\CurrentControlSet\Services\Spooler /v Start /d 0x4` and pressed Enter. This set the Start value to 0x4. Clicked Y to confirm the command.



```
Administrator: Command Prompt
C:\Windows\system32>reg add HKLM\SYSTEM\CurrentControlSet\Services\Spooler /v Start /d 0x4
Value Start exists, overwrite(Yes/No)? Y
The operation completed successfully.
C:\Windows\system32>
```

6. Typed command `reg import /?` To view the import help information.

```
C:\Windows\system32>reg import /?

REG IMPORT FileName [/reg:32 | /reg:64]

    FileName    The name of the disk file to import (local machine only).

    /reg:32     Specifies the key should be accessed using the 32-bit registry view.

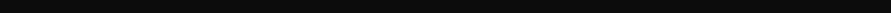
    /reg:64     Specifies the key should be accessed using the 64-bit registry view.

Examples:

    REG IMPORT AppBkUp.reg
    Imports registry entries from the file AppBkUp.reg

C:\Windows\system32>
```

7. Typed the command `reg import C:\SpoolerBak.reg` and pressed Enter.



The screenshot shows a Windows Command Prompt window titled "Administrator: Command Prompt". The command prompt displays the following text:

```
C:\Windows\system32>reg import C:\SpoolerBak.reg
The operation completed successfully.
C:\Windows\system32>
```

8. Typed command reg query HKLM\SYSTEM\CurrentControlSet\Services\Spooler pressed Enter and verified that the Start value is 0x2as it was restored from the backup.

```
C:\Windows\system32>reg query HKLM\SYSTEM\CurrentControlSet\Services\Spooler
```

```
HKKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Spooler
    DependOnService REG_MULTI_SZ RPCSS\0http
    Description     REG_SZ @%systemroot%\system32\spoolsv.exe,-2
    DisplayName     REG_SZ @%systemroot%\system32\spoolsv.exe,-1
    ErrorControl    REG_DWORD 0x1
    FailureActions   REG_BINARY 100E0000000000000000000000300000014000000010000008813000001000000881300000000000000
0000
    Group           REG_SZ SpoolerGroup
    ImagePath       REG_EXPAND_SZ %SystemRoot%\System32\spoolsv.exe
    ObjectName      REG_SZ LocalSystem
    RequiredPrivileges REG_MULTI_SZ SeTcbPrivilege\0SeImpersonatePrivilege\0SeAuditPrivilege\0SeChangeNotifyPrivil
ege\0SeAssignPrimaryTokenPrivilege\0SeLoadDriverPrivilege
    ServicesIdType REG_DWORD 0x1
    Start          REG_DWORD 0x2
    Type           REG_DWORD 0x110

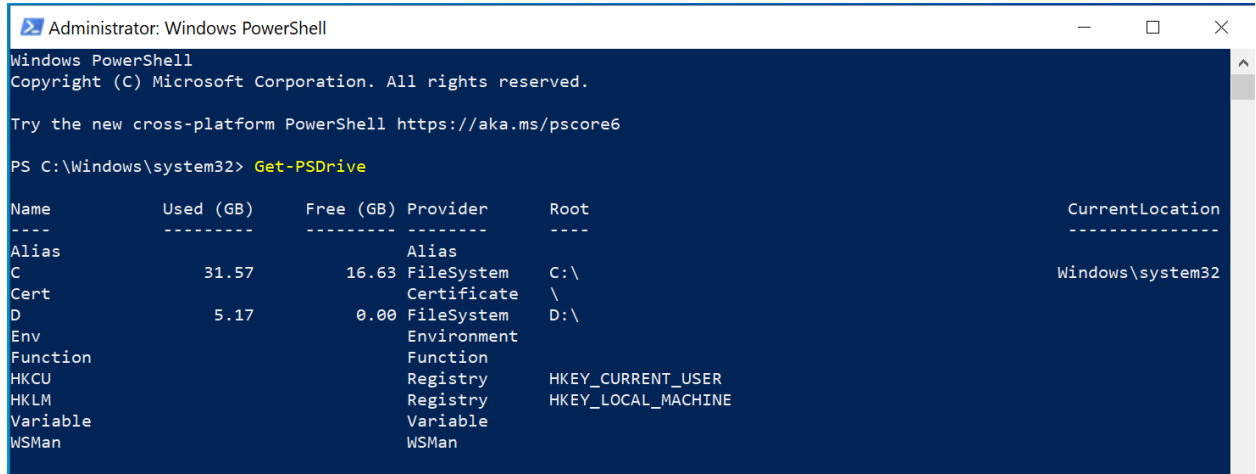
HKKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Spooler\Performance
HKKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Spooler\Security

C:\Windows\system32>
```

9. Closed the command prompt.

Activity 9-6: Viewing the Registry by Using Windows PowerShell

1. Opened the Powershell (Admin) and typed Get-PSDrive and pressed Enter to see if HKCU and HKLM are available.



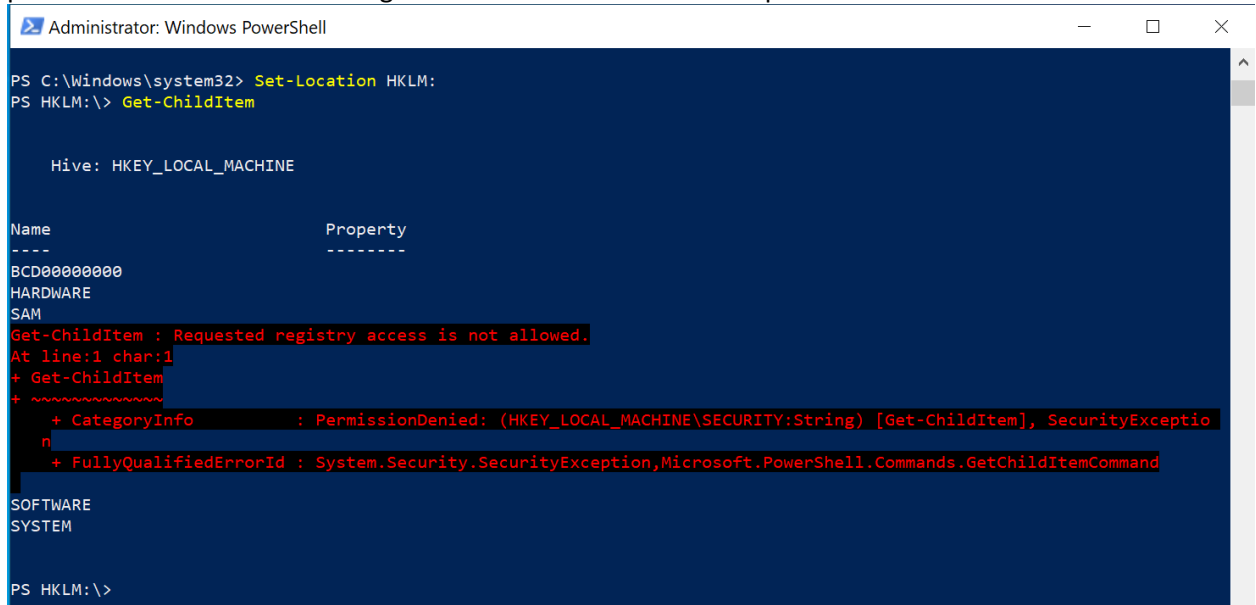
```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> Get-PSDrive
```

Name	Used (GB)	Free (GB)	Provider	Root	CurrentLocation
Alias			Alias		
C	31.57	16.63	FileSystem	C:\	Windows\system32
Cert			Certificate	\	
D	5.17	0.00	FileSystem	D:\	
Env			Environment		
Function			Function		
HKCU			Registry	HKEY_CURRENT_USER	
HKLM			Registry	HKEY_LOCAL_MACHINE	
Variable			Variable		
WSMan			WSMan		

2. Typed command Set-location HKLM: to enter HKLM prompt. Then Typed Get-ChildItems and pressed Enter. Got a Red message and continued to the next step.



```
Administrator: Windows PowerShell

PS C:\Windows\system32> Set-Location HKLM:
PS HKLM:\> Get-ChildItem

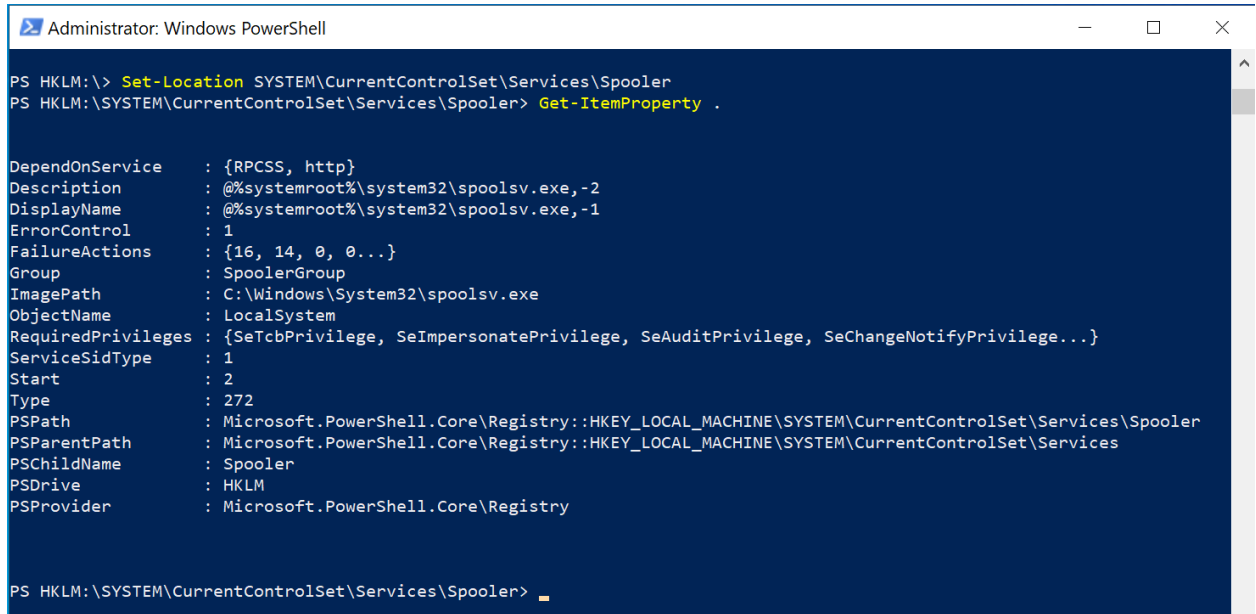
Hive: HKEY_LOCAL_MACHINE

Name                Property
----                -
BCD00000000
HARDWARE
SAM
Get-ChildItem : Requested registry access is not allowed.
At line:1 char:1
+ Get-ChildItem
+ ~~~~~
+ CategoryInfo          : PermissionDenied: (HKEY_LOCAL_MACHINE\SECURITY:String) [Get-ChildItem], SecurityExceptio
n
+ FullyQualifiedErrorId : System.Security.SecurityException,Microsoft.PowerShell.Commands.GetChildItemCommand

SOFTWARE
SYSTEM

PS HKLM:\>
```

3. Typed command `Set-Location SYSTEM\CurrentControlSet\Services\Spooler` and pressed enter to execute the command. Then typed `Get-ItemProperty .` command and noticed that Start has value 2.



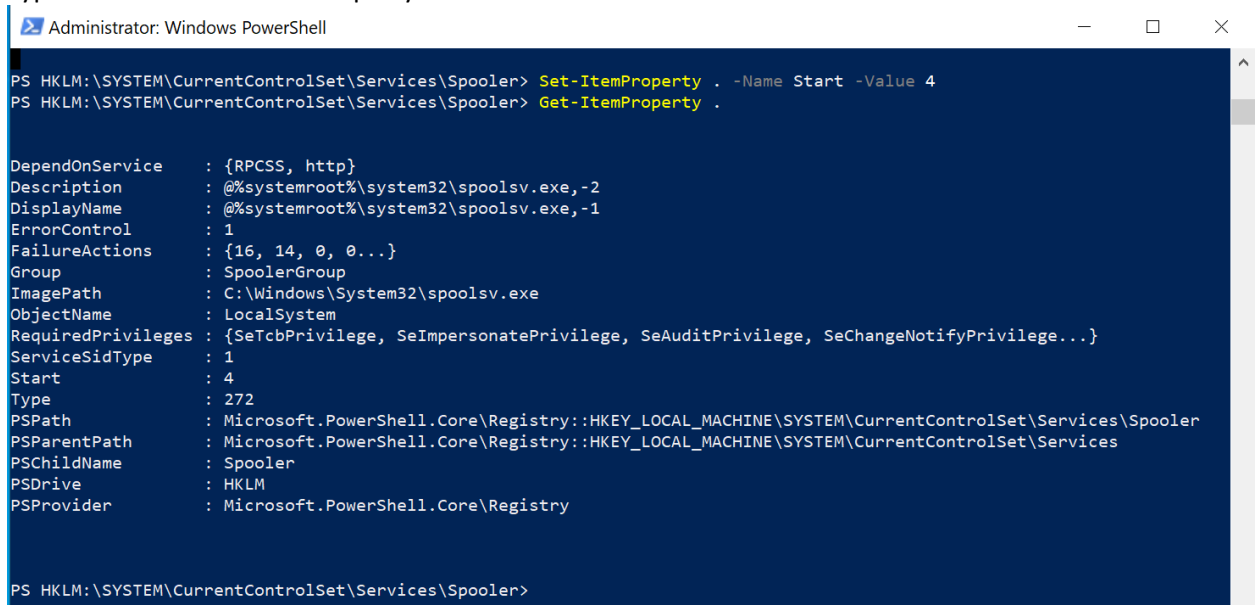
```
Administrator: Windows PowerShell

PS HKLM:\> Set-Location SYSTEM\CurrentControlSet\Services\Spooler
PS HKLM:\SYSTEM\CurrentControlSet\Services\Spooler> Get-ItemProperty .

DependOnService      : {RPCSS, http}
Description           : @%systemroot%\system32\spoolsv.exe,-2
DisplayName           : @%systemroot%\system32\spoolsv.exe,-1
ErrorControl          : 1
FailureActions        : {16, 14, 0, 0...}
Group                 : SpoolerGroup
ImagePath             : C:\Windows\System32\spoolsv.exe
ObjectName            : LocalSystem
RequiredPrivileges    : {SeTcbPrivilege, SeImpersonatePrivilege, SeAuditPrivilege, SeChangeNotifyPrivilege...}
ServiceSidType        : 1
Start                 : 2
Type                  : 272
PSPath                : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Spooler
PSParentPath          : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services
PSChildName           : Spooler
PSDrive               : HKLM
PSProvider             : Microsoft.PowerShell.Core\Registry

PS HKLM:\SYSTEM\CurrentControlSet\Services\Spooler>
```

4. Typed command `Set-ItemProperty . -Name Start -Value 4` to set the Start value 4.



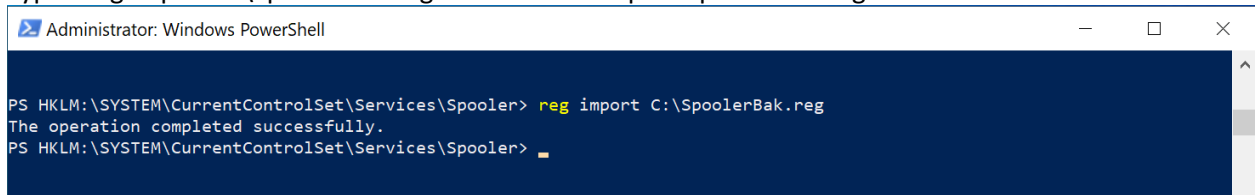
```
Administrator: Windows PowerShell

PS HKLM:\SYSTEM\CurrentControlSet\Services\Spooler> Set-ItemProperty . -Name Start -Value 4
PS HKLM:\SYSTEM\CurrentControlSet\Services\Spooler> Get-ItemProperty .

DependOnService      : {RPCSS, http}
Description           : @%systemroot%\system32\spoolsv.exe,-2
DisplayName           : @%systemroot%\system32\spoolsv.exe,-1
ErrorControl          : 1
FailureActions        : {16, 14, 0, 0...}
Group                 : SpoolerGroup
ImagePath             : C:\Windows\System32\spoolsv.exe
ObjectName            : LocalSystem
RequiredPrivileges    : {SeTcbPrivilege, SeImpersonatePrivilege, SeAuditPrivilege, SeChangeNotifyPrivilege...}
ServiceSidType        : 1
Start                 : 4
Type                  : 272
PSPath                : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Spooler
PSParentPath          : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services
PSChildName           : Spooler
PSDrive               : HKLM
PSProvider             : Microsoft.PowerShell.Core\Registry

PS HKLM:\SYSTEM\CurrentControlSet\Services\Spooler>
```

5. Typed `reg import C:\SpoolerBak.reg` command to import SpoolerBak.reg



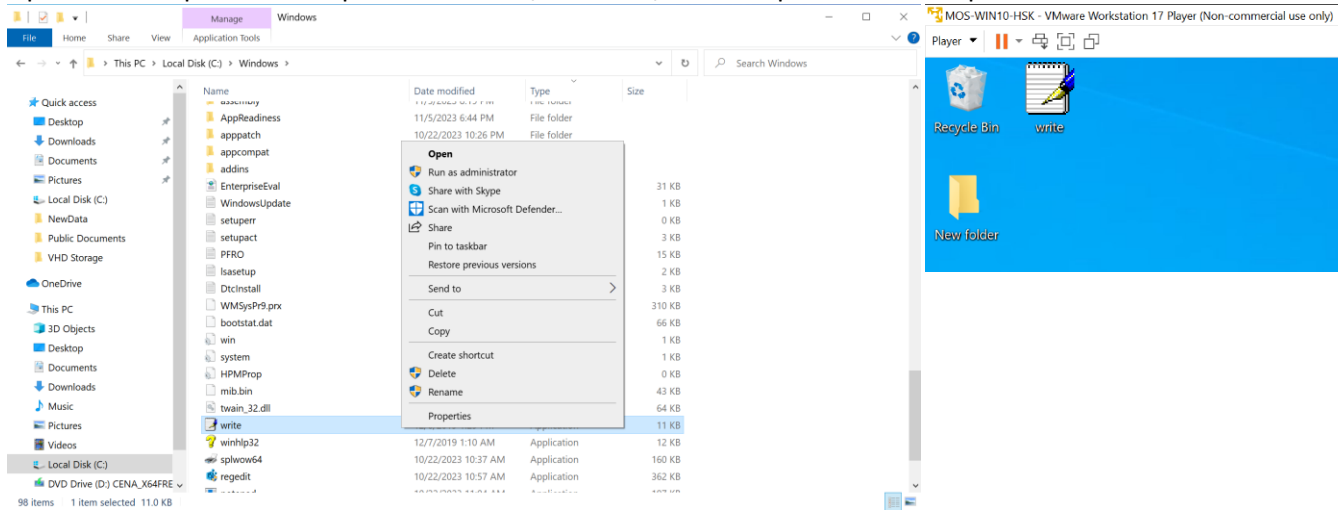
```
Administrator: Windows PowerShell

PS HKLM:\SYSTEM\CurrentControlSet\Services\Spooler> reg import C:\SpoolerBak.reg
The operation completed successfully.
PS HKLM:\SYSTEM\CurrentControlSet\Services\Spooler>
```

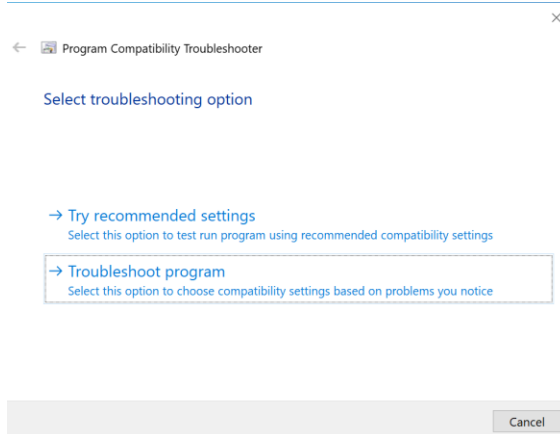
6. Closed the PowerShell prompt.

Activity 9-7: Configuring app Compatibility

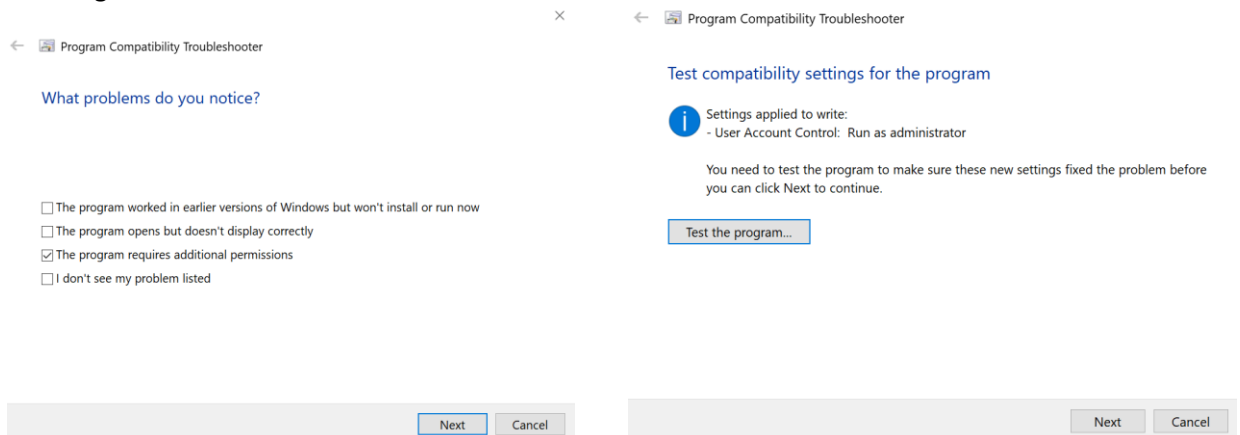
1. Opened File explorer and copied write at C:\Windows\write and paste on Desktop.



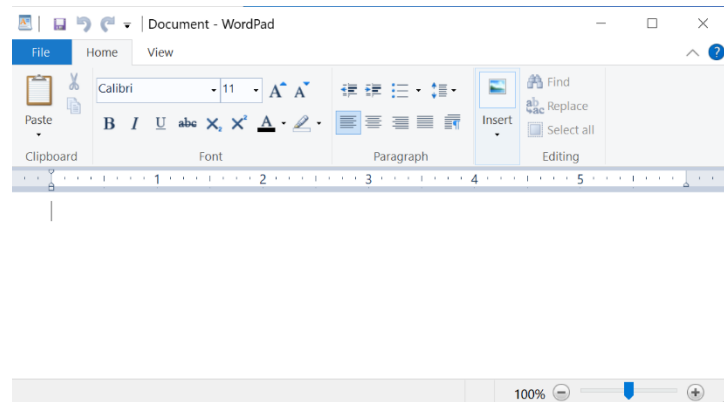
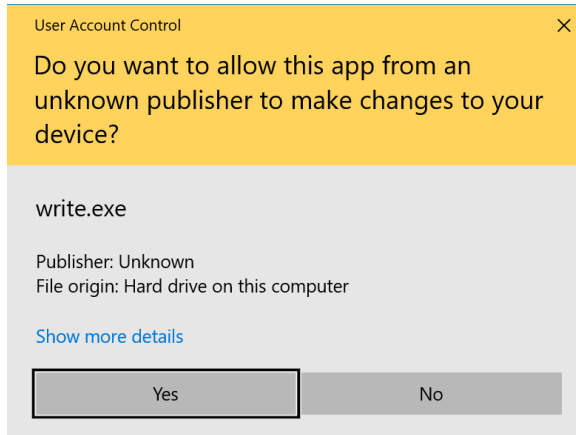
2. Right Clicked write and selected Troubleshoot compatibility and clicked Troubleshoot program on Program Compatibility Troubleshooter window.



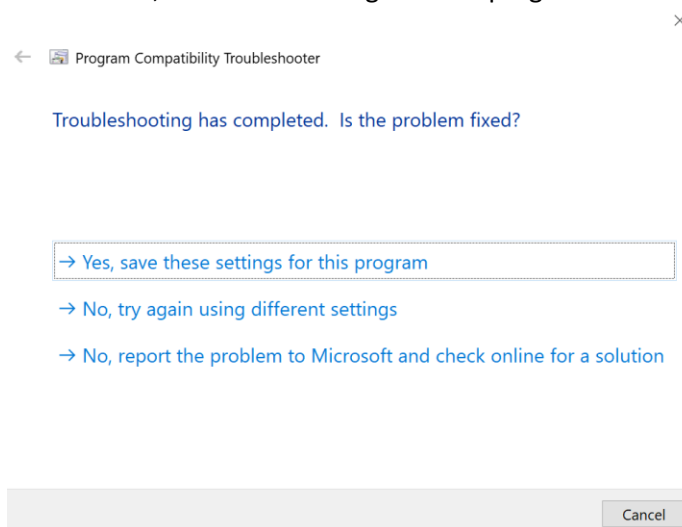
3. Selected option The Program requires additional permissions and clicked next. Then clicked Test the Program on next window.



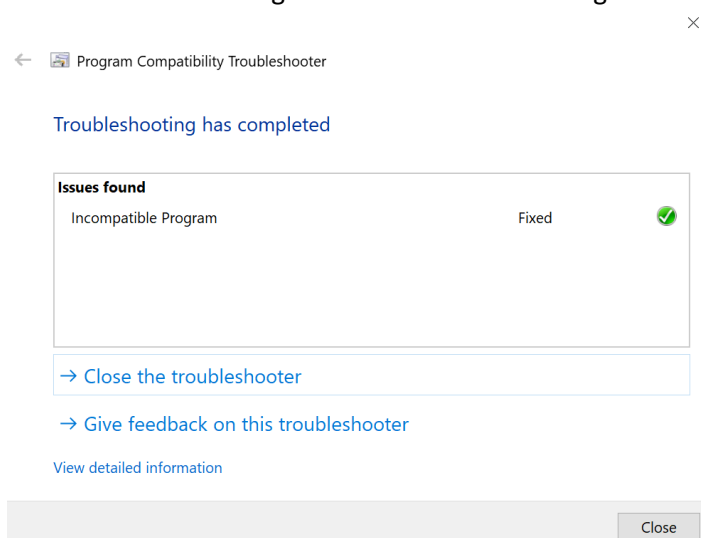
4. Clicked Yes on User account control dialog box and it opened a WordPad. Then closed the WordPad.



5. Selected Yes, save these settings for this program on the Next dialog box.



6. I received the following screen on Troubleshooting has completed window. Then I clicked close.



7. Right clicked on the write and opened properties window. Noticed that run this program as an administrator is selected and then clicked cancel.

